

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following state equations is/are linear, so could be used in a linear Kalman filter such as the one derived in the second course of this specialization? Select all that apply.

1 / 1 point

☐ $x_k = 2x_{k-1} + 0.5u_{k-1} + w_{k-1} + 3.$

☒ $x_k = (0.9)^{k-1}x_{k-1} + w_{k-1}.$

✔ Correct

Yes. This is a linear (time-varying) state equation with $A_k = (0.9)^k$ and $B_k = 0.$

☒ $x_k = 0.9x_{k-1} + 0.2u_{k-1}.$

✔ Correct

Yes. This is a linear state equation with $A_k = 0.9$ and $B_k = 0.2.$ Since there is no noise term, we also have that $\Sigma_{\bar{w}} = 0.$

☐ $x_k = 0.8x_{k-1} \times u_{k-1} + w_{k-1}.$

☐ $x_k = 0.9x_{k-1}^2 + 0.3u_{k-1} + w_{k-1}.$

2. Which types of state estimator assume in their derivations that all uncertain quantities (process noise, sensor noise, state estimate) have Gaussian distribution? Select all that apply.

1 / 1 point

☒ The linear Kalman filter.

✔ Correct

Yes. This filter is derived based on general Gaussian sequential probabilistic inference, tailored to a specific set of extra assumptions.

☒ The extended Kalman filter.

✔ Correct

Yes. This filter is derived based on general Gaussian sequential probabilistic inference, tailored to a specific set of extra assumptions.

☒ The sigma-point Kalman filter.

✔ Correct

Yes. This filter is derived based on general Gaussian sequential probabilistic inference, tailored to a specific set of extra assumptions.

☒ The unscented Kalman filter.

✔ Correct

Yes. This filter is derived based on general Gaussian sequential probabilistic inference, tailored to a specific set of extra assumptions.

☐ The particle filter.

3. Which steps from the linear Kalman filter must be modified by the extended and sigma-point Kalman filters to account for a nonlinear system model? Select all that apply.

1 / 1 point

☒ Step 1a: State estimate time update.

✔ Correct

Yes. This step must be modified since the expectation operation must be computed in a different way for a nonlinear system model.

☒ Step 1b: Error covariance time update.

✔ Correct

Yes. This step must be modified since the expectation operation must be computed in a different way for a nonlinear system model.

☒ 1c: Estimate system output.

✔ Correct

Yes. This step must be modified since the expectation operation must be computed in a different way for a nonlinear system model.

☒ 2a: Estimator gain matrix.

✔ Correct

Yes. This step must be modified since the expectation operations required to compute $\Sigma_{\hat{x}y,k}^-$ and $\Sigma_{\hat{y},k}^{-1}$ must be computed in a different way for a nonlinear system model.

☐ 2b: State estimate measurement update.

☐ 2c: Covariance estimate measurement update.